

Project Proposal: CivicBot

A Region-Based Multilingual Robotic Assistant for Community Engagement

1. Executive Summary

CivicBot is a region-assigned, multilingual robotic civic assistant that addresses delayed infrastructure maintenance and accessibility barriers in urban spaces, particularly for vulnerable groups such as the elderly, pupils, and tourists. Current reporting systems are often strictly digital, excluding individuals with low technological literacy. CivicBot operates as a permanent community support unit, where each robot is responsible for a specific geographic zone, ensuring localized knowledge, accountability, and trust. It combines public infrastructure reporting, community support, elderly assistance, pupil guidance, and tourist information services, transforming robots into scalable and interoperable components of modern civic infrastructure.

2. Description of the Robotic Solution:

CivicBot is a semi-autonomous mobile system deployed in neighborhoods, markets, and city centers. The model utilizes a fleet where each unit is assigned a defined region.

Core Functions:

Infrastructure Reporting: Autonomously detects and reports road damage, broken lighting, waste, and obstacles with visual evidence and location data.

Community Interaction: Provides a physical touch and voice interface, allowing citizens to report issues without smartphones or apps.

Vulnerable Group Support: Offers accessible guidance and service information for the elderly, and provides safety navigation and educational awareness for pupils.

Tourist & Language Aid: Acts as a multilingual guide offering directions, maps, and local information to non-native speakers.

3. Innovation and Uniqueness

CivicBot shifts focus from isolated functionality to system-level design. Its key innovations include:

Region-Based Responsibility: Permanent assignment to specific areas builds localized data and community trust.

Multifunctional Utility: It converges infrastructure monitoring, social assistance, and tourism guidance into one platform.

Inclusive Physical Interface: It democratizes smart city services, requiring no digital literacy or personal devices to use.

Scalability: The architecture allows expansion from a single neighborhood prototype to a city-wide network.

4. Impact on Society

Aligned with UN SDG 11 (Sustainable Cities), CivicBot fosters safer, more inclusive communities.

- **Social Impact:** It accelerates infrastructure repair, reduces language barriers, enhances independence for the elderly, and improves safety for students and pedestrians.
- **Beneficiaries:** Citizens, elderly individuals, school pupils, tourists, local authorities, and NGOs.

5. System Construction

The system is built on modular, upgradable technologies designed for reliability.

Hardware Components:

Mobile robotic base, high-resolution camera (visual monitoring), touchscreen/physical interface, onboard microcontroller/computer (edge processing), wireless communication module, and environmental sensors.

Software Components:

Navigation and motion control, Human–Robot Interaction (HRI) interface, image-based issue detection, multilingual interaction system, and a centralized backend reporting platform.

System Flow:

Sensing - Processing - Classification - Reporting/Assistance - Feedback

6. Key Design and Programming Ideas

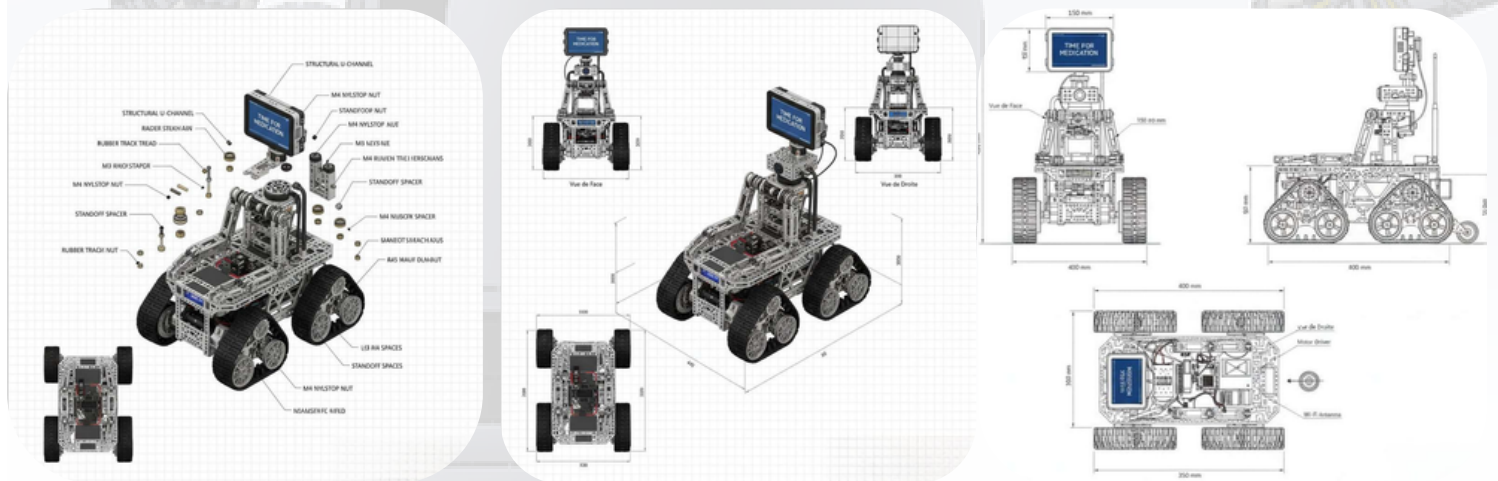
Modular Design: Hardware and software are independent for easy upgrades.

Region-Aware Logic: Operations are strictly bound by geographic coordinates and local data.

Edge Processing: Core processing occurs locally to reduce latency.

Event-Based Reporting: Issues are automatically prioritized and categorized.

Open Development: Built using open, well-documented technologies.



Conclusion

CivicBot is a practical, innovative solution that integrates community engagement with smart public-space management. By combining regional responsibility, multilingual accessibility, and a scalable fleet design, the project meets critical social needs while demonstrating technical robustness.